



Engineering Constitution & 23 Agent Instruction Briefs · Enterprise-grade build law



147 new pages of governance. A binding Engineering Constitution (12.3 pages, Ten Laws + 24 articles) plus a complete standing instruction brief for each of the 23 agent roles (5.5–6.4 pages each, 134.7 pages total). Written as a 25-year software engineer and architect would write them: rules that exist because someone once lost data, lost a wedding, or lost a customer.

The plan package now contains **72 files / ~194,000 words / ~353 pages**. Everything below is inside the zip at the bottom of this page.

What was added

File	What it is	Size
<code>12-ENGINEERING-CONSTITUTION.md</code>	The operating law: Ten Laws, 24 articles, enforcement lists	12.3 pages
<code>agents/AGENT-INDEX.md</code>	Roster, discipline groupings, every standing veto	1.5 pages
<code>agents/AGENT-<CODE>-<ROLE>.md</code> × 23	One complete instruction brief per role	134.7 pages
<code>README.md</code>	Now indexes the governance layer and the reading order	updated

The Ten Laws

Absolute. Not trade-offs to be balanced against schedule. Every pull request is judged against them.

1. **The photographer's originals are read-only, forever.** Touching an original is a Severity 1 incident.
2. **Manual work is sacred.** `user_edited_fields` is checked before every write, in every code path.
3. **Determinism is a feature.** Same input, same models, same version, same output. Nondeterminism is a defect.
4. **Offline is the default, not the fallback.** Proven by a network-disabled CI job on every commit.
5. **Never fail silently.** No `unwrap`, no `expect`, no empty catch, no default substituted for a failure.
6. **Every automated decision is explainable and reversible.** One sentence, one click to override.
7. **No unbudgeted work.** Budgets measured on three named machines, asserted in CI, 5 % regression fails the build.
8. **Every long operation is cancellable and resumable.** 250 ms cancel; `kill -9` safe at any moment.
9. **Licence-clean or it does not ship.** Research-only assets are physically quarantined.
10. **Privacy is structural.** No client images off-device without consent. No file paths in logs — paths contain client names.

Two corollaries with equal weight: small, reviewable, reversible changes (<400 lines, feature-flagged); and **correct, then beautiful, then fast — in that order, always.**

Constitution article map

Article	Covers
Preamble	The seven properties that actually make software enterprise-grade

Article	Covers
I	The Ten Laws
II	Architecture: boundaries, dependency rule, ports and adapters, contracts before code, one writer per domain
III	Code rules for Rust, TypeScript and Python, including the no-panic and no- any regime
IV	Determinism and reproducibility: seeds, version stamping, reproducible builds
V	Data: forward-only migrations, provenance, recomputable state, backup and restore, PII scoping
VI	AI/ML: model cards, calibration, fairness per skin-tone bucket, heuristic baselines, no silent model swaps, governed cloud budget
VII	Testing: the pyramid, zero flake tolerance, fixture weddings, property-based tests, never weaken an assertion
VIII	Performance: budgets are tests, three reference machines, distributions not averages, the 5 % rule
IX	Security and privacy: keychain-only secrets, default-deny egress, SBOM, vulnerability SLAs
X	Reliability: 99.5 % crash-free, journalling, kill -9 harness, idempotency, degrade never collapse
XI	Observability: structured logs, trace IDs, the decision ledger, redacted support bundles
XII	Version control: trunk-based, conventional commits, <400-line PRs, CODEOWNERS
XIII	CI/CD: mechanical gates, rollback before rollout, staged rollout, signing, feature flags
XIV	Change management and ADRs
XV	Definition of Ready (6 checks) and Definition of Done (12 checks)
XVI	Documentation rules
XVII	Compliance, licensing and legal hygiene
XVIII	Incident management: severities, mitigate-first, blameless postmortems
XIX	Accessibility and internationalisation
XX	Technical debt policy: 20 % of every phase reserved

Article	Covers
XXI	Metrics: DORA plus product metrics, and deleting metrics nobody acts on
XXII	Governance, decision rights, standing vetoes, amendment process
XXIII	Rules for AI coding agents — the twelve operating rules for Claude Code
XXIV	Enforcement: what blocks a merge, what blocks a release, how waivers work

The 23 agent briefs

Each brief has 17 sections: how to use it, ownership map, phase assignments, non-negotiable rules, SOP per phase, interfaces and handoffs, quality gates owned, definition of done, anti-patterns, decision rights and veto, toolbox, first week, how you are measured, the universal law, the PR checklist, the escalation protocol, and a closing word from 25 years of experience.

Code	Role	Reports to	Veto	Pages
CTO	Chief Architect / CTO	user	yes	6.4
PM	Product Manager	user	yes	6.3
EM	Engineering Manager / Delivery Lead	CTO	yes	6.2
TLC	Tech Lead, Imaging Core (Rust)	CTO	yes	6.1
SRC	Senior Engineer, Core Pipeline	TLC	yes	5.8
SRG	Senior Engineer, GPU and Render	TLC	yes	5.9
MLL	ML Lead, Vision	CTO	yes	6.3
SRML	Senior ML Engineer	MLL	no	5.7
MLR	ML Research Engineer	MLL	no	5.7

Code	Role	Reports to	Veto	Pages
MLOPS	MLOps / Model Packaging	MLL	yes	5.8
AGT	AI Agent and Prompt Engineer	CTO	yes	6.0
COL	Colour Scientist	CTO	absolute	5.9
SFE	Senior Frontend Engineer	EM	yes	5.8
MFE	Mid-Level Frontend Engineer	SFE	no	5.5
MBE	Mid-Level Backend / Cloud Engineer	EM	no	5.6
DATA	Data Engineer / Dataset Curator	MLL	yes	5.7
QAL	QA Lead, Automation	EM	yes	5.7
QAIQ	QA Engineer, Image Quality	QAL	yes	5.8
PERF	Performance Engineer	CTO	yes	5.7
DEVOPS	DevOps / Release Engineer	EM	yes	5.7
SEC	Security and Privacy Engineer	CTO	absolute	5.8
UX	UX / UI Designer	PM	yes	5.8
DOC	Technical Writer	PM	yes	5.7

Standing vetoes

A veto is technical, written, and **cannot be overruled by schedule pressure** — only satisfied.

- **COL** — skin and colour rendering, and render node order. Absolute.
- **SEC** — secrets, network egress, licence compatibility, client data. Absolute.

- **QAIQ** — perceptual regressions: plastic skin, halos, banding, face and hand artefacts, gallery inconsistency.
 - **QAL** — missing or flaky tests, incomplete release verification matrix.
 - **PERF** — unjustified performance-budget regressions.
 - **DEVOPS** — unsigned artefacts, untested rollback.
 - **DATA** — unlicensed training data, wedding-level split leakage.
 - **TLC** — non-cancellable, non-resumable or corrupting changes.
 - **MLL / MLOPS** — models without a card, calibration, fallback or signature.
 - **CTO** — dependency cycles, the offline guarantee, unversioned formats, destructive edits.
 - **PM** — overclaiming, and automation without an override.
 - **UX** — automated decisions shown without a reason and a reversible override.
 - **DOC** — shipping undocumented user-facing behaviour.
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Definition of Done — all twelve, every time

- ☐ Acceptance criteria demonstrably met, with evidence attached
- ☐ Tests at the right layer, including negative and interruption paths
- ☐ All phase quality gates green on all three reference machines
- ☐ Performance budgets met and asserted in CI
- ☐ No new `unwrap`, `expect`, `panic!`, `any` or swallowed error
- ☐ Errors typed and actionable; all four interface states handled
- ☐ Determinism verified: two runs, identical output
- ☐ Cancellation and resume verified for any long operation
- ☐ Documentation and glossary updated in the same pull request
- ☐ Telemetry and ledger entries added where decisions are made
- ☐ Handoff note written with contracts and measured numbers

☐ Feature flag added if risky, and rollback stated

Enforcement

-  **Blocks a merge:** lint or `clippy::pedantic` failure · any new `unwrap` / `expect` / `panic!` / `any` · missing or flaky tests · a weakened assertion · a red quality gate · golden-image drift · >5 % performance regression without written justification · licence or vulnerability finding · a new network destination · schema change without a forward-only migration · missing documentation, handoff note or model card · a PR over 400 lines · **any violation of the Ten Laws, for which there is no exception process.**
-  **Blocks a release:** incomplete verification matrix · any known flake · any perceptual regression from QAIQ · crash-free below 99.5 % · failed 24-hour soak · untested rollback · unsigned or unnotarised artefact · unpinned `models.lock` · a fixture wedding that will not complete offline · known critical or high vulnerability · unpublished SBOM · undocumented user-facing behaviour · a red budget entry without an approved justification.

There is exactly one exception mechanism: a **written waiver** naming the rule, the reason, the risk, the expiry date and the approving veto-holder, recorded in the repository. No verbal exceptions. Waivers expire; they do not become custom.

Rules for AI coding agents (Article XXIII)

This is the article that most directly governs how you drive Claude Code:

1. Read `CLAUDE.md`, the constitution, the phase file and your agent brief **before** producing code.
2. **One phase at a time, one concern at a time.** Never let an agent implement three phases in one session.

3. **Contracts first, in a separate commit** — this is what lets several agents work without collision.
4. Never invent an interface that already exists. Search the repository first.
5. Tests are part of the deliverable, not a follow-up task.
6. State assumptions explicitly and **stop when blocked**. Never guess a threshold, a licence or a benchmark.
7. **Never fabricate a measurement**. Numbers come from a run, never from a plausible estimate.
8. Respect the invariants absolutely, however convenient a violation would be.
9. Keep changes under 400 lines. If the task is bigger, split it and say so.
10. Write the handoff note. This is how a 23-agent team stays coherent across months.
11. Prefer deleting to adding. Complexity an agent adds, a human must maintain.
12. When rules conflict, escalate rather than choose.

Reading order for every Claude Code session

1. `CLAUDE.md`
2. `12-ENGINEERING-CONSTITUTION.md`
3. `agents/AGENT-<YOUR-ROLE>.md`
4. `phases/PHASE-NN-<SLUG>.md` — **one phase only**

Complete package

[AURA-Wedding-AI-Master-Plan.zip](#)

[AURA-Wedding-AI-Master-Plan.zip](#) · 72 files · 30 phases · constitution · 23 agent briefs